Francesco Mascari

he/him | PhD Student

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francesco-mascari

Education

PhD in Statistics and Computer Science — **2021 – ongoing**
Bocconi University, Milan, Italy

MSc in Mathematics for Life and Data Sciences, cum laude — **2018 – 2021**
University of Trento, Trento, Italy

BSc in Mathematics, cum laude — **2015 – 2018**
University of Bologna, Bologna, Italy

High School Diploma in Scientific Studies, cum laude — **2010 – 2015**
Liceo Scientifico Copernico, Bologna, Italy

MSc Thesis

Title: *A Micro-macro Connection: The Valuable Relation between Large Deviations for Diffusion Processes and Wasserstein Gradient Flows*

Supervisors: Dr. Carlo Orrieri, *University of Trento*; Dr. Giovanni Conforti, *École Polytechnique*

Abstract: This work aims to present the passage from the microscopic stochastic description of a system of particles to its macroscopic description as a Fokker-Planck equation. After a brief introduction to some basics in Convex Analysis and Measure Theory, a complete review of optimal transport and Wasserstein metric, on the one hand, and of large deviations for path measures of diffusion processes, on the other, is displayed. A thorough description of the gradient flow formulation for the Fokker-Planck equation is then analyzed with a particular interest in the JKO functional, which is shown to be equivalent in Γ-convergence to the rate functional arising in the large deviation principle for a system of independent Brownian particles.

Exchange and Visiting Programs

Erasmus+ Traineeship — **Oct 2020 – Feb 2021**
École Polytechnique, Palaiseau, France

Erasmus+ Study — **Jan – Jun 2018**
Cardiff University, Cardiff, Wales, UK

Conferences, Workshops and Summer Schools

2023 LMS Invited Lecture Series — **Jul 2023**
Durham University, Durham, England, UK

Turin-Bath PhD Workshop in Applied Probability and Statistics — **Jun 2023**
Collegio Carlo Alberto, Turin, Italy

Workshop Mathematical Statistics — **Apr 2023**
Bocconi University, Milan, Italy

Optimal Transport and Applications — **Oct 2022**
University of Pisa, Pisa, Italy

Summer School in Mathematics of Machine Learning — **Jul 2022**
Scuola Matematica Interuniversitaria, Cortona, Italy

Summer School in Advanced Statistics and Probability: Random Structures and Combinatorial Statistics — **Jul 2022**
Lake Como School of Advanced Studies, Como, Italy

Stochastic Games and Martingale Optimal Transport — **May 2022**
University of Milan, Milan, Italy

Summer School in Geometric Statistics — **Sep 2019**
University of Toulouse III, Toulouse, France

Teaching

Teaching Assistantships

Statistics - (30001) — **Fall 2023**
BSc International Economics and Finance, Bocconi University

Bayesian Statistical Methods - (20231) — **Fall 2023**
MSc Data Science and Business Analytics, Bocconi University

Mastering Data for Insurance — **Spring 2023**
SDA Bocconi

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Optimization - (20603) **Spring 2023**
MSc Data Science and Business Analytics, Bocconi University

Mathematics - module 2 (applied) - (30063) **Spring 2023**
BSc International Economics and Finance, Bocconi University

Advanced Analysis and Optimization - module 1 (30551) **Fall 2022**
BSc Mathematical and Computing Sciences for Artificial Intelligence, Bocconi University

Mathematics and Statistics II - (145105) **Spring 2020**
BSc Biomolecular Sciences and Tecnologies, University of Trento

Calculus A - (145503) **Fall 2019**
BSc Mathematics, University of Trento

Calculus I - (145432) **Fall 2019**
BSc Physics, University of Trento

Other teaching activities

Math Help Desk **Spring 2020**
Applicant refugee students and asylum seeker, University of Trento

Volunteering

Member of the PhD Board **2022 – ongoing**
Department of Decision Sciences, Bocconi University, Milan, Italy

Member of the Department Council **2016 – 2018**
Department of Mathematics, University of Bologna, Bologna, Italy

Member of the School Board **2013 – 2014**
Liceo Scientifico Copernico, Bologna, Italy

Skills

- **Coding** (R, Python, MATLAB, Julia): Advanced
- **Office suite** (Word, Excel, PowerPoint, OneDrive): Advanced
  ECDL Full Standard issued in Apr 2015.
- **Data management** (Office Access, SQL) : Basic
- **Data analysis and machine learning** (PCA, k-means, regression models, bayesian networks) : Intermediate
  Machine Learning course by Stanford University @ Coursera issued in May 2021.
  Probabilistic Graphical Model specialization by Stanford University @ Coursera issued in Apr 2023.
- **Soft skills**: Communication (public speaking, team working); organization (time-management, adaptability).
  BEAT PhD Students – Bocconi Excellence in Advanced Teaching by Bocconi University issued in Nov 2022.

Languages

- **Italian:** Native proficiency
- **English:** C1 – Full professional working proficiency
  IELTS Academic 7.0 issued in Jul 2016.
  Cambridge FCE issued in Jan 2015.
- **French:** A2 – Elementary proficiency

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